

Converting simple microemulsion into attractive foam for topical pain treatment. A case study with lidocaine base.

E. Dauphin-Chanard; C. André ; D. Hoste ; D. Marchaud
GATTEFOSSÉ – Saint-Priest – France – edauphin@gattefosse.com

PURPOSE

Lidocaine base is a local anesthetic agent. It belongs to the WHO Model List of Essential Medicines, the most important medications needed in a basic healthcare system¹. Lidocaine base is used to prevent and reduce pain or discomfort caused by skin irritations². The study aims to develop microemulsion-based foams with lidocaine base, to match practicality and efficiency. Microemulsions increase drug solubility and drug permeation through the skin³. These liquid dispersions of oil and water phases with a surfactant/co-surfactant system are thermo-dynamically stable. They differ from conventional emulsions by their physical properties: transparency, low viscosity and small particle size (< 200 nm). A screening of surfactant, co-surfactant and oily vehicle was investigated using pseudo-ternary diagrams to determine the microemulsion areas providing foaming capacity. Propellant-free pump allows foam production by introducing air in the liquid microemulsion⁴.

METHOD

Materials

Caprylocaproyl macrogol-8 glycerides (Labrasol®), as surfactant, polyglyceryl-3 dioleate (Plurol® Oleique CC 497), oleoyl polyoxylglycerides (Labrafil® M 1944 CS), lauroyl macrogol-6 glycerides (Labrafil® M 2130 CS), as co-surfactant, medium chain triglycerides (Labrafac™ lipophile WL 1349), as oily vehicle, were provided by Gattefosse (France). Polysorbate 80 (Tween®80) is a surfactant supplied by BASF (Germany). Lidocaine base (Moebs Iberica, Spain) was used as model drug. Propellant-free pump devices (50ml, PET) were purchased from Packit (Germany).

Method

Evaluation of solubility of lidocaine base

The solubility of lidocaine base was determined at 25°C after two weeks in each individual excipient. Drug concentration was determined by HPLC (Waters chromatograph Alliance 2695D, coupled with a 2487 model UV detector at 254 nm).

Design of microemulsion zones in pseudo-ternary diagram

Pseudo-ternary phase diagrams were constructed by titration of homogenous liquid mixtures of oil, surfactant, and co-surfactant, with water at room temperature to determine the microemulsion areas. 4% W/W lidocaine base was then added to a selection of microemulsions.

Microemulsion characterization

pH, droplet size (Dynamic Light Scattering – DLS – method) were measured and microscopic observations carried out.

Foamability evaluation of microemulsion

Liquid microemulsions were tested in a propellant-free pump for their capacity to generate foam. The visual assessment was done over time (instantly and after 3 minutes).

CONCLUSION

This study highlights the foaming capacity of microemulsion containing Labrasol®. Its dual function as surfactant and powerful solubilizer enables the realization of foam containing 4% of lidocaine base.

This study underlines the potential of using microemulsion in propellant-free device to obtain foams.

Benefits for the patients are transparency, solubilizing properties and efficiency of microemulsion combined with practicality, easy application and good spreadability. Moreover industrial advantages include simple process, easy scale-up, low cost device without propellant gas. Pharmaceutical foams keep its promise on the topical market.

RESULTS AND DISCUSSION

Selection of surfactant (S)

The solubility of lidocaine base was determined in individual surfactant, co-surfactant, oil and water. Among the selected surfactants, Labrasol® is the best solubilizer of lidocaine base with solubility value around 360 mg/ml (versus 226 mg/ml for Tween® 80). All co-surfactants tested have similar solubilizing capacity: with 312 mg/ml, 310 mg/ml and 293 mg/ml for Plurol® Oleique CC497, Labrafil® M1944CS and Labrafil® M2130 CS respectively. **Labrasol® and Plurol® Oleique CC 497 have been selected as they are the best solubilizers for lidocaine base.**

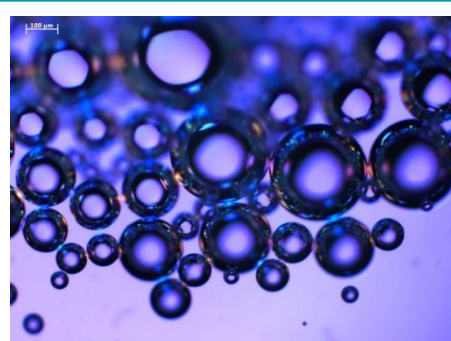
Design of microemulsion zones in pseudo-ternary diagram

Labrasol® / Plurol® Oleique CC 497 (6/1), Labrafac™ Lipophile WL 1349 and water produce microemulsion/ emulsion zones as shown on the pseudo-ternary diagram (figure 1). Several microemulsions (A, B, C, D and E) have been evaluated for their foaming capacity. The tests show that microemulsions should contain 15-30% of surfactant/cosurfactant and minimum 70% of water to produce foams of good quality and 3 minutes-persistency. 4% (W/W) of lidocaine base was added to this microemulsion. The optimal formula is composed of: 4% lidocaine base, 24.0% Labrasol®, 4.0% Plurol® Oleique CC497, 0.5% Labrafac™ Lipophile WL 1349, 66.5% demineralized water and 1% benzylic alcohol. This microemulsion provides foam with good quality and consistency. The DLS assessment shows small droplets (120 nm) that validates the microemulsion form. Microscopic evaluation of the optimized microemulsion confirms that lidocaine base is fully solubilized, as no crystals are visible (Table 1) even when displayed into foam.

Microscopy



Microemulsion (Magn. x20)



Microemulsion Foam (Magn. x10)

Macroscopy

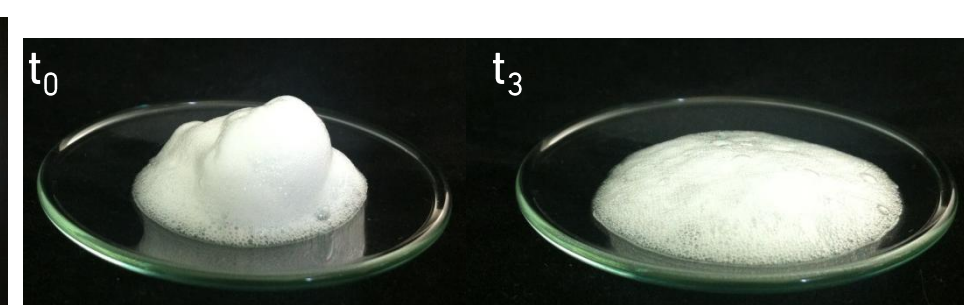
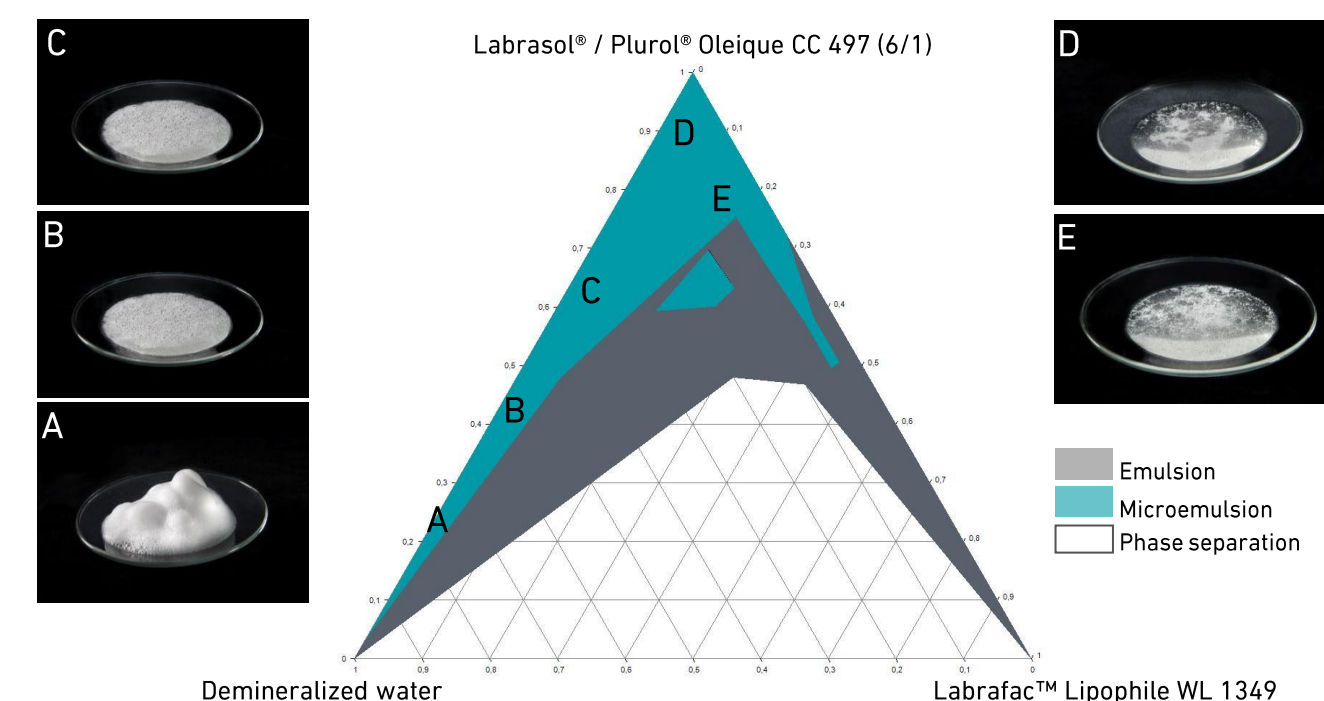


Figure 1. Microemulsion/emulsion zones



Microemulsion composition	A	B	C	D	E
Labrasol®	18.0%	35.0%	55.0%	70.0%	60.0%
Plurol® Oleique CC 497	3.0%	7.0%	11.0%	14.0%	12.0%
Labrafac™ Lipophile WL 1349	0.5%	3.0%	6.0%	9.0%	26.0%
Demineralized water	78.5%	55.0%	28.0%	7.0%	2.0%

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